

TiLiA – TimeLineAnnotator

Felipe D. Martins ^{1*}, Anne R. E. Foo ^{2*}, and Mark R. H. Gotham ^{3*}

¹ Universidade Federal do Rio de Janeiro, Brazil ² École Polytechnique Fédérale de Lausanne, Digital and Cognitive Musicology Lab, Switzerland ³ King's College London, United Kingdom * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#))

Summary

'TiLiA' (TimeLine Annotator) is an open-source, cross-platform network of tools and resources designed to facilitate real-time creating, displaying and interacting with timeline-style annotations of audio, video and musical scores. Developed in Python with the PySide UI framework, TiLiA is available primarily as a desktop Graphical User Interface (GUI), supported by a Command Line Interface (CLI) as well as a free and open online platform where users may upload, view and query existing analyses.

Statement of need

Visual analyses of music have proven to be an enduringly popular, effective, and versatile tool for conveying a variety of musical ideas.¹ As the famous adage goes: 'a picture is worth a thousand words'. These analytical images are primarily connected to the *analysis* rather than the *music*, thus making them readily useful not only for the already visual sources of notated music, but also for the much broader range of musical styles and repertoires that are transmitted without the use of notation.

A comprehensive analysis of music requires being able to summarise the relationships between events that are happening concurrently and through time. Furthermore, whatever potential a *static* image has, an *interactive* one offers considerably more. Formal analysis is one of the clearest beneficiaries of the visual summary of music. Form can be hard to parse (especially in real-time) but is often easy to understand in relatively simple, at-a-glance summaries.

In today's digital age, we have come to expect certain basic interactive features across visual interfaces, such as resizing content to fit the window, keyboard shortcuts, undo/redo functionality, and so on. Musical contexts extend this with additional expectations, such as **real-time synchronisation** (a.k.a., audio or score following). Any tool that aims to win over users effectively must present analytical ideas naturally and intuitively, with minimal friction in the user interface.

Save for a relatively small early body of literature (see the following section), musical scholarship has not yet made the leap into this technological space with any degree of decisiveness or consistency. Even most visually minded music analyses are consigned to static images embedded in print (or in e-copies at best) and not subject to the expectations for *fair*, open-source data sharing that are now *de rigueur* in other fields. Still among the more computationally inclined communities, there has been little progress made on the coordination of standards and corpora for formal analysis.

¹See Isaacson ([2023](#)) for a survey of this long history.

State of the field

While one *can* appropriate tools (whose primary purpose is editing scores or audio) for creating visual analyses, the current range of tools dedicated to the task typically offer limited functionality and their uptake by the wider community has been minimal.

Existing tools fall into a few groups, none of which meets this need on its own. General-purpose audio and notation software can be repurposed for analysis (Peebles, 2013), but it is built for editing rather than annotating: the timeline in *Sibelius*, for instance, offers only bar numbers and rudimentary checkpoints. Audio-analysis tools such as Sonic Visualiser (Cannam et al., 2010) and Partiels (Guillot, 2025) centre on audio and on features extracted by plug-ins, and lack integration with musical scores. Score-annotation platforms such as *Edirom Online* (Röwenstrunk, 2014), *Listen Here!* (Weigl et al., 2023) and *mei-friend* (Goebel & Weigl, 2024) are not specific to analysis, and the MPM Toolbox (Berndt, 2021) is tied to a single performance-markup format. Other timeline-based tools are either closed-source, like *Audio Timeliner* (Notess & Swan, 2004; Yorgason, 2018) and *iAnalyse* (Couprie, 2019) (which is also not cross-platform), or web-based, like the open-source *Dezrann* (Giraud et al., 2018). These do not have dedicated tools to annotate other musical relations beyond hierarchical structures.

Research systems show what such interfaces can offer. SyncPlayer (Kurth et al., 2005), the Walküre interface (Balke et al., 2017) and the Erkomaishvili interface (Rosenzweig et al., 2020) give synchronised access to specific corpora. Hierarchical structural annotations exist in datasets such as SALAMI (Smith et al., 2011), the Wagner Ring Dataset (Weiß et al., 2023) and in Gotham & Ireland (2019), alongside research on hierarchical segmentation (McFee & Ellis, 2014). These resources, however, come with their own interfaces or none at all. To our knowledge, no open-source, cross-platform tool combines alignment with audio, video, YouTube streams, PDFs and scores, dedicated musical annotation systems, and scriptable import and export.

In response to these challenges, we present ‘TiLiA’: a timeline annotator for all.

Specifications

TiLiA consists of an organisation of digital tools, primarily of its [open-source desktop application](#) and [command-line interface](#), but also of a supporting online platform. The desktop application is developed in Python using the PySide binding for the Qt UI framework. Automated testing is done with pytest, and fully-automated deployment is available via GitHub Actions. The code base is loosely organised around a Model-View-Controller (MVC) pattern, which allows both the CLI and the GUI to benefit from the same backend logic. We provide object-oriented base classes such as Timeline, TimelineUI and TimelineComponent as a means to promote extensibility and support an evergrowing range of annotation needs.

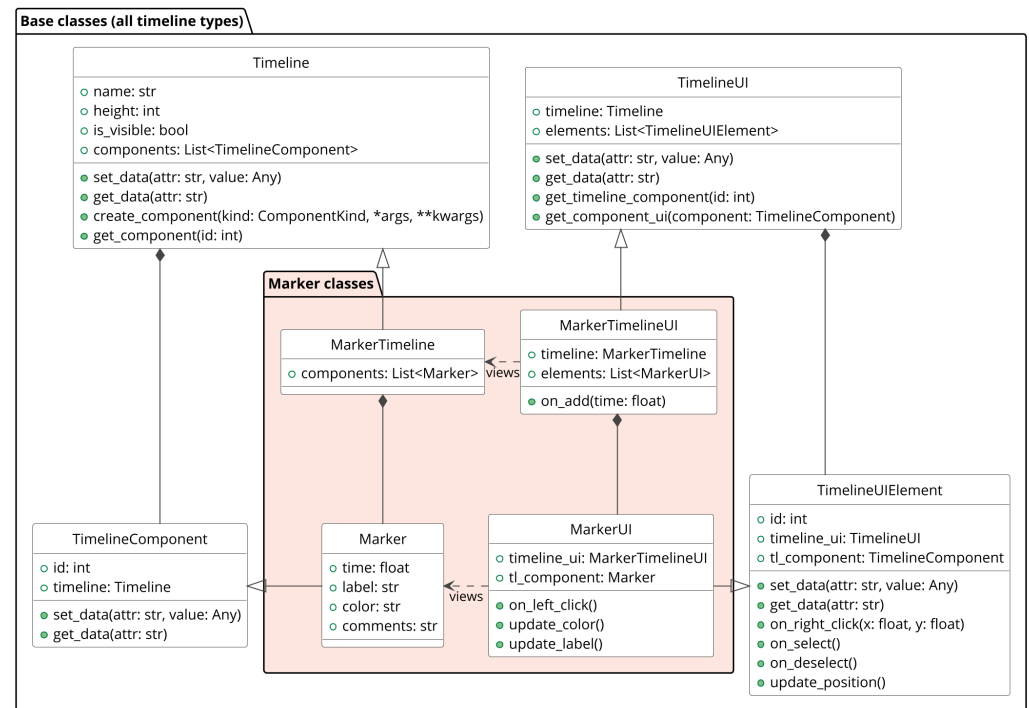


Figure 1: A simplified diagram for the main timeline base classes.

Figure 1 is a simplified diagram for the main timeline base classes. The outer box shows the base classes shared by all timeline types; the inner box shows the concrete subclasses for the marker timeline type, one example among the various timeline types. Timeline acts as container for TimelineComponents. Both act as models of TimelineUI and TimelineUIElement, which are views; in the marker example, MarkerTimelineUI and MarkerUI view MarkerTimeline and Marker respectively.

TiLiA is designed to facilitate the *creation* of analyses — particularly where this requires complex alignment with audio/video sources — and the (conversion and) *import* of analyses of external origin through CSV files. This effort thus connects TiLiA to wider datasets and scholarship, while also expanding its functionality. The [tilia-dcml](#) repository demonstrates how the TiLiA CLI might be used to visualise data from an external corpus of musical analyses. The CLI can also *export* timelines created in TiLiA to a JSON file suitable for processing by other software.

The desktop application

The main part of TiLiA is a cross-platform *What-You-See-Is-What-You-Get* GUI (Figure 2), designed to feel intuitive to anyone who has worked with audiovisual editing software. A TiLiA analysis consists of timelines, each containing annotations that can be aligned to an audio, video, YouTube stream or musical score. Each *timeline type* is designed to support conventional formal annotations, with the affordance to work with a variety of other analytical styles. These timelines each contain one or more *component types* (e.g., markers for marker timelines), each with different *properties* to convey information including comments, colour and labels.

We connect timelines and their components by using the loaded media as a “base layer” to which all timelines refer. Timeline components have either time or start_time and end_time timestamps which position them in relation to the underlying media. If a different media is loaded, the user has the option to either uniformly scale timelines to the new media or to keep



102 Currently, there are eight types of timelines:

Timeline type	Purpose
Audiowave	Displaying amplitude graphs
Beat	Marking beats/measures, synchronising metrical and absolute time
Harmony	Chord symbols and roman numeral analysis
Hierarchy	Representing hierarchical relations and sequences
Marker	Annotating discrete timepoints
PDF	Synchronising external PDF documents with playback
Range	Representing non-hierarchical relations and sequences
Score	Synchronising digitised scores with playback

103 Command-line interface

In addition to the desktop GUI and the web app, a **command line interface** allows the automation of file and timeline creation and edition, media loading and external data importing. Sequences of commands can be loaded by the CLI from scripts written in plain text files, enabling programmatic conversion from various data formats to TiLiA files.

```
108 metadata set-media-length 120
109 timelines add hierarchy --name "Form"
110 timelines add marker --name "Formal closures"
111 timelines add hierarchy --name "Harmonic segments"
112 timelines add beat --name "Measures" --beat-pattern 1
113 timelines add marker --name "Harmony"
114 timelines add marker --name "Harm. functions"
115 timelines add marker --name "Keys"
116
117 load-media <path to audio> --scale-timelines yes
```

```

118 metadata set title "Title"
119 save <path to TiLiA file> --overwrite
120 export <path to JSON file> --overwrite

```

121 Online platform

122 The TiLiA desktop application is supported by an [online platform](#) that lets users store, share,
 123 visualise and even query existing analyses. Registration is free and is only required for uploading
 124 content. This reflects a goal of TiLiA to support not only the modernised creation and storage
 125 of analyses, but also to support the still-rare practice of analytical *collaboration* and structured
 126 *searches* on the corpora; both which benefit from network effects facilitated by the lowered
 127 entry barrier.

128 Outlook

129 The TiLiA desktop application aims to provide a user-friendly annotation tool both for MIR
 130 researchers and musicologists. It can be used to create large annotation datasets or detailed,
 131 interactive analyses of individual pieces. It can also be used in the classroom to better
 132 connect theoretical concepts with audiovisual material or as a common interface for analytical
 133 assignments.

134 In the future we hope to convert existing open-source datasets and integrate them into the
 135 web platform, providing a centralised hub where diversified audiovisual data can be queried
 136 and interacted with in a uniform manner. We also plan to improve audio processing and score
 137 displaying capabilities and to add more timeline types (e.g. a 2-dimensional graph timeline) to
 138 offer more diversified visualisations.

139 References

- 140 Balke, S., Hiemer, M., Schwab, P., Arifi-Müller, V., Meyer-Wegener, K., & Müller, M. (2017).
 141 Die Oper als Multimediaszenario – Wagners Walküren gehen online. *INFORMATIK 2017*,
 142 75–86. https://doi.org/10.18420/in2017_04
- 143 Berndt, A. (2021). The Music Performance Markup Format and Ecosystem. *Proceedings*
 144 *of the 22nd International Society for Music Information Retrieval Conference*. [https://](https://archives.ismir.net/ismir2021/paper/000005.pdf)
 145 archives.ismir.net/ismir2021/paper/000005.pdf
- 146 Cannam, C., Landone, C., & Sandler, M. (2010). Sonic Visualiser: An Open Source Application
 147 for Viewing, Analysing, and Annotating Music Audio Files. *Proceedings of the 18th ACM*
 148 *International Conference on Multimedia*. <https://doi.org/10.1145/1873951.1874248>
- 149 Couprie, P. (2019). iAnalyse 5 : Le développement d'un logiciel pour la musicologie numérique.
 150 *Journées d'informatique Musicale (JIM 2019)*. <https://hal.science/hal-02133538>
- 151 Giraud, M., Groult, R., & Leguy, E. (2018). Dezzrann, a Web Framework to Share Music
 152 Analysis. In S. Bhagwati & J. Bresson (Eds.), *Proceedings of the International Conference*
 153 *on Technologies for Music Notation and Representation – TENOR'18* (pp. 104–110).
 154 Concordia University. ISBN: 978-1-5251-0551-7
- 155 Goebl, W., & Weigl, D. M. (2024). Mei-friend: An interactive web-based editor for digital
 156 music encodings. *Journal of Open Source Software*, 9(98), 6002. [https://doi.org/10.](https://doi.org/10.21105/joss.06002)
 157 [21105/joss.06002](https://doi.org/10.21105/joss.06002)
- 158 Gotham, M. R. H., & Ireland, M. (2019). Taking Form: A Representation Standard, Conversion
 159 Code, and Example Corpora for Recording, Visualizing, and Studying Analyses of Musical
 160 Form. In A. Flexer, G. Peeters, J. Urbano, & A. Volk (Eds.), *Proceedings of the 20th*
 161 *International Society for Music Information Retrieval Conference, ISMIR 2019, Delft, The*

- 162 Netherlands, November 4–8, 2019 (pp. 693–699). [http://archives.ismir.net/ismir2019/](http://archives.ismir.net/ismir2019/paper/000084.pdf)
163 [paper/000084.pdf](http://archives.ismir.net/ismir2019/paper/000084.pdf)
- 164 Guillot, P. (2025). Partiels – Exploring, Analyzing and Understanding Sounds. *Proceedings*
165 *of the 28th International Conference on Digital Audio Effects (DAFx25)*, 298–303. [https://](https://www.dafx.de/paper-archive/2025/DAFx25_paper_66.pdf)
166 www.dafx.de/paper-archive/2025/DAFx25_paper_66.pdf
- 167 Isaacson, E. (2023). *Visualizing Music*. Indiana University Press. [https://www.barnesandnoble.](https://www.barnesandnoble.com/w/visualizing-music-eric-isacson/1140906228?ean=9780253064738)
168 [com/w/visualizing-music-eric-isacson/1140906228?ean=9780253064738](https://www.barnesandnoble.com/w/visualizing-music-eric-isacson/1140906228?ean=9780253064738)
- 169 Kurth, F., Müller, M., Damm, D., Fremerey, C., Ribbrock, A., & Clausen, M. (2005).
170 SyncPlayer – An Advanced System for Multimodal Music Access. *Proceedings of the 6th*
171 *International Conference on Music Information Retrieval*. [https://ismir2005.ismir.net/](https://ismir2005.ismir.net/proceedings/1025.pdf)
172 [proceedings/1025.pdf](https://ismir2005.ismir.net/proceedings/1025.pdf)
- 173 McFee, B., & Ellis, D. P. W. (2014). Analyzing song structure with spectral clustering.
174 *Proceedings of the 15th International Society for Music Information Retrieval Conference*.
175 <https://archives.ismir.net/ismir2014/paper/000319.pdf>
- 176 Notess, M., & Swan, M. (2004). Timeliner: Building a learning tool into a digital music library.
177 In L. Cantoni & C. McLoughlin (Eds.), *Proceedings of EdMedia + innovate learning 2004*
178 (pp. 603–609). Association for the Advancement of Computing in Education (AACE).
179 <https://www.learntechlib.org/p/12995>
- 180 Peebles, C. (2013). Using Audacity to Participate in Active Musical Listening. *Engaging*
181 *Students: Essays in Music Pedagogy*. <http://flipcamp.org/engagingstudents/peebles.html>
- 182 Rosenzweig, S., Scherbaum, F., Shugliashvili, D., Arifi-Müller, V., & Müller, M. (2020).
183 Erkomaishvili Dataset: A Curated Corpus of Traditional Georgian Vocal Music for
184 Computational Musicology. *Transactions of the International Society for Music Information*
185 *Retrieval*, 3(1), 31–41. <https://doi.org/10.5334/tismir.44>
- 186 Rösenstrunk, D. (2014). *Markieren, Codieren, Annotieren – Arbeiten an und mit digitalen*
187 *Musikeditionen mit Edirom und MEI*.
- 188 Smith, J. B. L., Burgoyne, J. A., Fujinaga, I., De Roure, D., & Downie, J. S. (2011). Design
189 and creation of a large-scale database of structural annotations. *Proceedings of the 12th*
190 *International Society for Music Information Retrieval Conference*, 555–560.
- 191 Weigl, D. M., VanderHart, C., Rammler, D., Pescoller, M., & Goebel, W. (2023). Listen
192 here! A web-native digital musicology environment for machine-assisted close listening.
193 *Proceedings of the 10th International Conference on Digital Libraries for Musicology*,
194 109–118. <https://doi.org/10.1145/3625135.3625144>
- 195 Weiß, C., Arifi-Müller, V., Krause, M., Zalkow, F., Klauk, S., Kleinertz, R., & Müller, M.
196 (2023). Wagner Ring Dataset: A Complex Opera Scenario for Music Processing and
197 Computational Musicology. *Transactions of the International Society for Music Information*
198 *Retrieval*, 6(1), 135–149. <https://doi.org/10.5334/tismir.161>
- 199 Yorgason, B. (2018). Teaching with Timelines. *Engaging Students*, 6. [http://flipcamp.org/](http://flipcamp.org/engagingstudents6/essays/yorgason.html)
200 [engagingstudents6/essays/yorgason.html](http://flipcamp.org/engagingstudents6/essays/yorgason.html)